(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 2058 Roll No.

# B.Tech

## (SEM V) ODD SEMESTER THEORY EXAMINATION 2009-10

## COMMUNICATION ENGG.

*Time : 3 Hours]* *[Total Marks : 100*

***Note :*** *Attempt all questions.*

**1** Attempt any **four** parts of the following : **5×4=20**

(a) A voltage

$$V = 200\left(1 + 0.4\sin 2\pi f_m t\right)\sin 2\pi f_c t$$

is applied to a resistor of 100 ohms. Find the power dissipated by each of the frequency components presents in the voltage V.

(b) Explain frequency division multiplexing (FDM).

(c) A carrier wave of a frequency of 20 kHz is amplitude modulated signal

$f(t) = \cos 2\pi 10^3 t + \cos 4\pi 10^3 t$. Find the expression for the corresponding SSB-SC signal.

(d) Explain how the constant intermediate frequency is achieved in the superheterodyne receiver.

(e) What is simple automatic gain control ? What are its function ?

(f) Explain and draw block diagram of an SSB transmitter using the filter system.

2 Attempt any **four** parts of the following : **5×4=20**

(a) Determine and draw the instantaneous frequency of a wave having a total phase angle given by

$$\psi(t) = 2000t + \sin 10t$$

(b) What is the function of the balanced modulator in the Armstrong modulation system ?

(c) Show that the output signal to noise ratio in an FM system is related to the AM system as follows :

$$\frac{[S_o/N_o]_{FM}}{[S_o/N_o]_{AM}} = 3\left(\frac{AK_f}{w_m}\right)^2$$

(d) Show that a low pass filter can be used as a discriminator.

(e) Explain spike generation and threshold effect in FM.

(f) Explain the effect of random noise on the output of an FM receiver fitted with an amplitude limiter.

3 Attempt any two parts of the following : **10×2=20**

(a) What is quantization error ? How does it depend upon the step size ? Suggest some methods to overcome the difficulties encountered when the modulating signal amplitude swing is large.

(b) Describe delta modulation systems. What are its limitations ? How can they be overcome ?

(c) Explain adaptive Delta modulation. And compare between PCM and DM.



4 Attempt any **two** parts of the following : **10×2=20**

(a) Describe in detail the main abnormal ionospheric variations, including a brief mention of the interference that may be caused by the sporadic E layer.

(b) Briefly explain :

(i) Stationary satellite system

(ii) Transponders.

(c) Explain in brief :

(i) Cellular system

(ii) Personal communication system.

5 Attempt any **two** parts of the following : **10×2=20**

(a) Using circuit diagram, explain how sync pulses are obtained from the composite video waveform, and how, in turn horizontal sync pulses are extracted.

(b) briefly discuss :

(i) Evanescent field

(ii) Cross Hanchen shift

(iii) Mode coupling.

(c) Define normalised frequency for an optical fiber and explain its uses in determination of number of guided modes propagating within a step index fiber.

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